CUSTOMER BEHAVIOR ANALYTICS

Turning Retail Data into Actionable Business Insights

Prepared by: Youssef Talaat
Project Type: End-to-End Data Analytics
Tools: Python, SQL, PostgreSQL, Power BI
Business Focus: Customer Insights, Product Performance, Discount Effectiveness, and Retail Decision Support

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1. Executive Summary

This project analyzes consumer shopping behavior to identify purchasing patterns, evaluate product performance, understand customer segments, and investigate the relationship between discounts, shipping options, subscription status, and purchase amounts.

The analysis covers 3,900 purchase records and combines data preparation, database analysis, and interactive business reporting to translate raw data into actionable insights.

Key Results

METRIC	RESULT
TOTAL PURCHASE RECORDS	3,900
TOTAL RECORDED PURCHASE VALUE	$233,081
AVERAGE PURCHASE AMOUNT	$59.76
AVERAGE CUSTOMER REVIEW RATING	3.75/5
DISCOUNTED PURCHASE RECORDS	1,677
OVERALL DISCOUNT RATE	43.0%
MISSING REVIEW RATINGS IDENTIFIED	37

Key Business Findings

- Male-associated records account for $157,890 in recorded purchase value, compared with $75,191 for female-associated records.
- Clothing accounts for 44.54% of purchase records, followed by Accessories at 31.79%.
- Young Adults contribute the highest recorded purchase value among the four age groups, at $62,143.
- Express-shipping transactions have a higher average purchase amount than Standard-shipping transactions.
- The overall discount rate is 43.0%, making promotional effectiveness an important area for further investigation.
- Several products have relatively high discount rates and may warrant a review of promotional strategy.
- Customer segmentation and repeat-purchase analysis provide a foundation for developing targeted retention initiatives.

Some subscription-related results and the discount-by-gender visualization require further validation before they can support management decisions.

Business Value

The project demonstrates how retail data can be transformed into structured business insights that support inventory planning, customer engagement, promotional evaluation, and performance monitoring.

The results identify opportunities for improvement; they do not claim that any revenue or profit increase has already been achieved.

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2. Business Problem and Objectives

2.1 Business Problem

A retail company wants to better understand its customers' shopping behavior to improve sales performance, customer satisfaction, and long-term loyalty.

Management needs to understand how purchasing patterns vary across demographics and product categories, and whether discounts, customer ratings, seasons, shipping options, and subscription status are associated with purchasing behavior.

Without reliable analysis, decisions about promotions, inventory, and customer engagement may depend on assumptions rather than evidence.

2.2 Project Objectives

The project aims to:

1. Understand customer purchasing patterns and demographics.
2. Identify the product categories and products with the highest purchase-record volumes.
3. Analyze discount utilization and differences in purchase amounts.
4. Compare purchasing behavior across shipping and subscription groups.
5. Segment records according to previous purchase activity.
6. Build an interactive Power BI dashboard for business reporting.
7. Translate findings into practical recommendations and measurable KPIs.

2.3 Business Questions

The analysis addresses the following questions:

- How does recorded purchase value vary by gender and age group?

- Which product categories account for the largest share of purchase records?
- Which products have the highest average customer ratings?
- How does average purchase amount differ by shipping type?
- Which products have the highest discount rates?
- How many discounted purchases meet or exceed the overall average purchase amount?
- How do purchase amounts differ by subscription status?
- How can purchasing history support customer segmentation?
- Which products rank highest within their categories?
- How are repeat-purchase records distributed across subscription groups?
- How do seasonal purchasing patterns differ?

The dataset does not contain a clearly identified sales-channel field, so online-versus-offline performance cannot be reliably evaluated using the available data.

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3. Dataset Overview

The analysis uses the Customer Shopping Behavior dataset, containing 3,900 records and 18 original columns.

The data includes customer demographics, products, purchase amounts, locations, seasons, review ratings, subscription status, shipping types, discount usage, payment methods, and previous purchasing activity.

Main Data Categories

CATEGORY	EXAMPLES
CUSTOMER INFORMATION	Age, gender, location
PRODUCT INFORMATION	Product purchased, category, size, color
PURCHASE INFORMATION	Purchase amount, season, purchase frequency
CUSTOMER FEEDBACK	Review rating
CUSTOMER RELATIONSHIP	Subscription status, previous purchases
TRANSACTION PREFERENCES	Shipping type, payment method, discount status

Two analytical features were added during data preparation: age group and estimated purchase-frequency days. A redundant promotional-code field was removed after a consistency check.

The final prepared dataset is expected to contain 19 columns.

Figure 1 — Dataset Overview



```
df.head()
```

[3]

Python

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Venmo	Fortnightly
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	Cash	Fortnightly
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Credit Card	Weekly
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	PayPal	Weekly
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	PayPal	Annually

Caption: Overview of the retail shopping dataset and its main analytical fields.

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4. Data Quality and Preparation

Data quality was reviewed before conducting the business analysis.

4.1 Missing Values

The initial inspection identified 37 missing customer review ratings.

These values were filled using the median rating within the corresponding product category. This approach preserves the affected records and accounts for potential differences in rating distributions between categories.

The missing-value count should be verified after the treatment.

4.2 Standardization

Column names were standardized to improve consistency across the analytical workflow.

4.3 Feature Engineering

Two additional analytical features were created:

- **Age Group:** Divides the records into Young Adult, Adult, Middle-aged, and Senior groups using age quartiles.
- **Purchase Frequency in Days:** Converts frequency categories into approximate day intervals to make comparisons easier.

The age groups are dataset-based analytical segments, while the frequency values are approximations rather than exact observed intervals between transactions.

4.4 Redundant Field Check

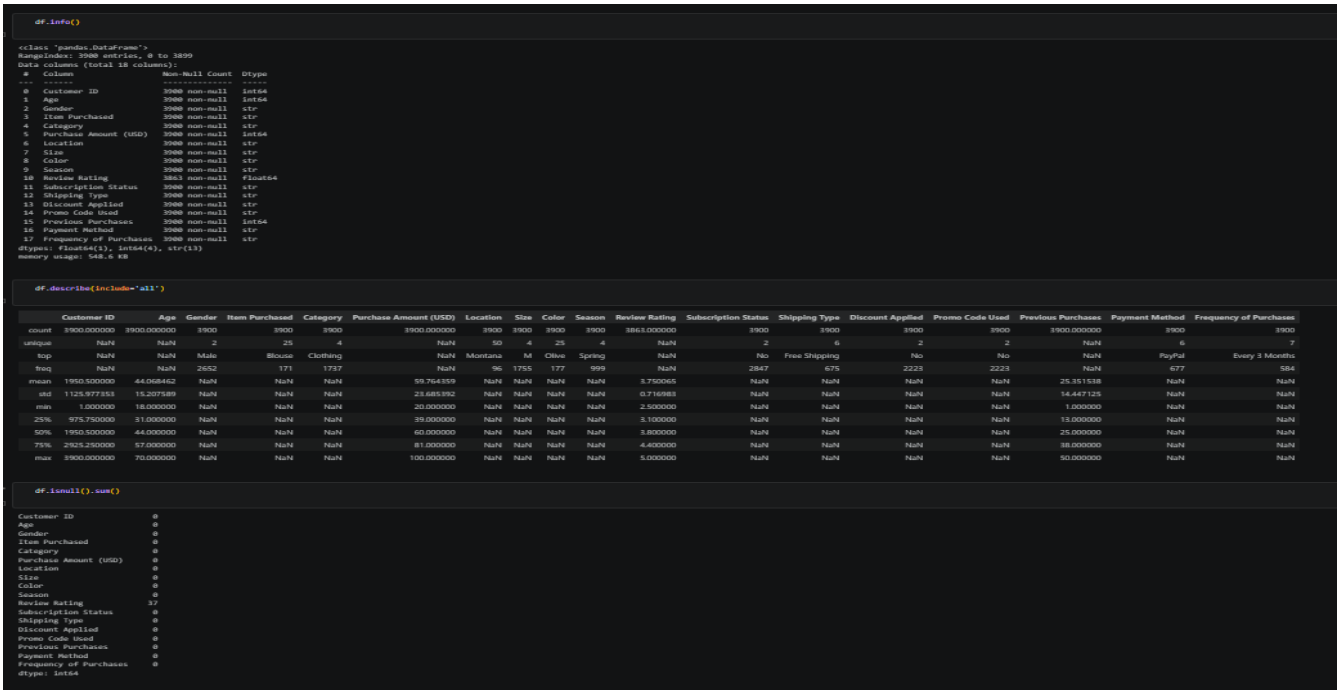
The relationship between discount status and promotional-code usage was checked. The two fields matched in the reported validation, so the redundant field was removed from the prepared dataset.

4.5 Quality Assurance Priorities

The final review should confirm:

- The expected number of records and columns.
- The treatment of missing review ratings.
- Consistency between SQL results and Power BI measures.
- Correct discount and subscription classifications.
- Whether customer identifiers represent unique customers or repeated records.

Figure 2 — Data Quality



```
df.info()

<class 'pandas.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Customer ID             3900 non-null   int64
 1   Age                     3900 non-null   int64
 2   Gender                  3900 non-null   str
 3   Item Purchased          3900 non-null   str
 4   Category                3900 non-null   str
 5   Purchase Amount (USD)   3900 non-null   int64
 6   Location                3900 non-null   str
 7   Size                    3900 non-null   str
 8   Color                   3900 non-null   str
 9   Season                  3900 non-null   str
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   str
 12  Shipping Type           3900 non-null   str
 13  Discount Applied        3900 non-null   str
 14  Promo Code Used         3900 non-null   str
 15  Previous Purchases      3900 non-null   int64
 16  Payment Method          3900 non-null   str
 17  Frequency of Purchases  3900 non-null   str
dtypes: float64(1), int64(4), str(13)
memory usage: 548.6 KB
```

```
df.describe(include='all')
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900	3900.000000	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2	NaN	6	7
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No	NaN	PayPal	Every 3 Months
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223	NaN	677	584
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN	25.351538	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN	14.447125	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN	1.000000	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN	13.000000	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN	25.000000	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN	38.000000	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN	50.000000	NaN	NaN

```
df.isnull().sum()

Customer ID               0
Age                       0
Gender                    0
Item Purchased            0
Category                  0
Purchase Amount (USD)     0
Location                  0
Size                      0
Color                     0
Season                    0
Review Rating            37
Subscription Status       0
Shipping Type             0
Discount Applied          0
Promo Code Used           0
Previous Purchases        0
Payment Method            0
Frequency of Purchases    0
dtype: int64
```

Caption: Data quality assessment and preparation before analysis.

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5. Analytical Approach

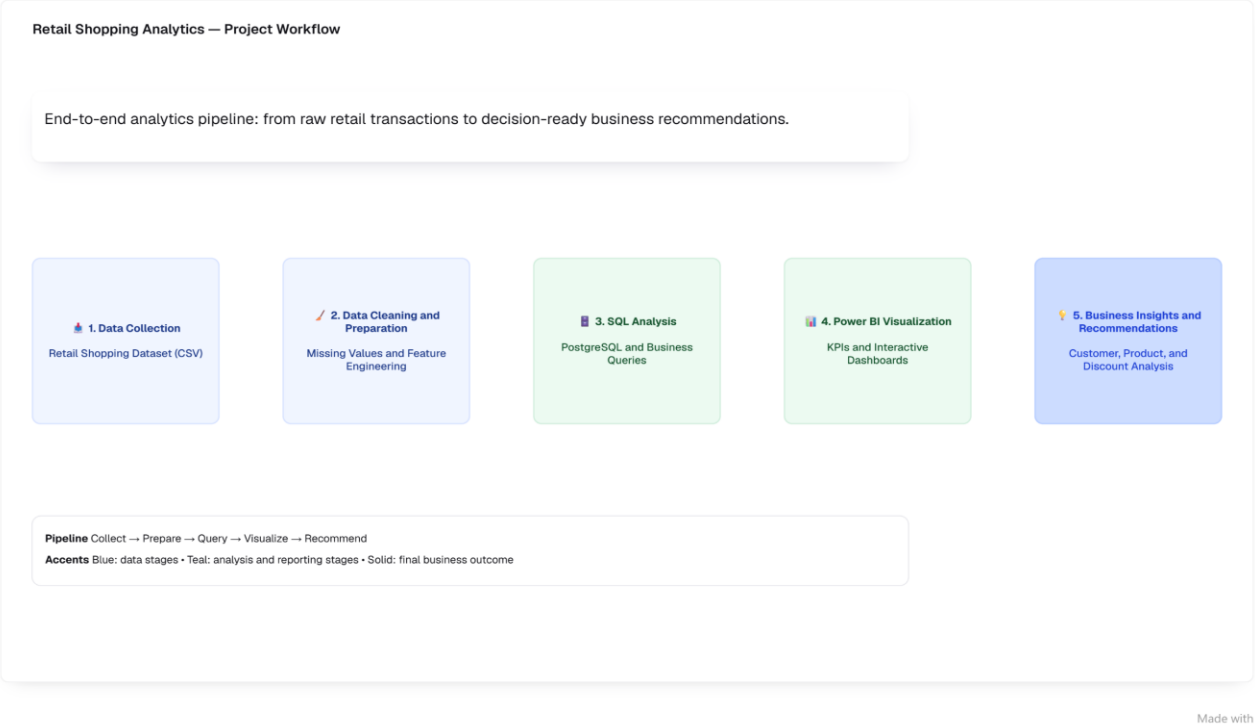
The project follows an end-to-end analytics workflow:

Dataset → Data Preparation → PostgreSQL Database → SQL Analysis → Power BI Dashboard → Business Recommendations

Each stage contributes to the final business outcome:

- **Python:** Inspects, cleans, and prepares the dataset.
- **PostgreSQL:** Stores the prepared data in a structured database.
- **SQL:** Answers business questions through aggregation, filtering, ranking, and segmentation.
- **Power BI:** Presents KPIs and purchasing patterns through interactive visualizations.
- **Business Analysis:** Converts the findings into recommendations and measurable next steps.

Figure 3 — Analytics Workflow



Caption: End-to-end workflow used to transform raw retail data into business insights.

Figure 4 — Database Verification

Data Output | Messages | Notifications

Showing rows: 1 to 10 | Page No: 1 of 1

	customer_id bigint	age bigint	gender text	item_purchased text	category text	purchase_amount bigint	location text	size text	color text	season text	review_rating double precision	subscription_status text	shipping_type text
1	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express
2	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express
3	3	50	Male	Jeans	Clothing	73	Massachu...	S	Maroon	Spring	3.1	Yes	Free Shipping
4	4	21	Male	Sandals	Footwear	90	Rhode Isla...	M	Maroon	Spring	3.5	Yes	Next Day Air
5	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoi...	Spring	2.7	Yes	Free Shipping
6	6	46	Male	Sneakers	Footwear	20	Wyoming	M	White	Summer	2.9	Yes	Standard
7	7	63	Male	Shirt	Clothing	85	Montana	M	Gray	Fall	3.2	Yes	Free Shipping
8	8	27	Male	Shorts	Clothing	34	Louisiana	L	Charcoal	Winter	3.2	Yes	Free Shipping
9	9	26	Male	Coat	Outerwear	97	West Virgi...	L	Silver	Summer	2.6	Yes	Express
10	10	57	Male	Handbag	Accessori...	31	Missouri	M	Pink	Spring	4.8	Yes	2-Day Shipping

Total rows: 10 | Query complete 00:00:00.075 | CRLF | Ln 1, Col 32

Caption: Verification of the prepared dataset in the PostgreSQL database.

6. SQL Analysis and Business Questions

6.1 Revenue by Gender

Results

GENDER	RECORDED PURCHASE VALUE	SHARE OF TOTAL
MALE	$157,890	67.74%
FEMALE	$75,191	32.26%
TOTAL	**$233,081**	**100.00%**

Business Insight

Male-associated records account for approximately two-thirds of the total recorded purchase value.

This difference may reflect variations in purchase frequency, product preferences, transaction counts, or average purchase amounts. The totals alone do not establish the underlying cause.

Recommendation

Compare average purchase amount, purchase-record count, product-category preferences, and purchase frequency across the groups before making marketing-allocation decisions.

6.2 High-Value Discounted Purchases

Result

839 purchase records had a discount applied and a purchase amount equal to or greater than the overall average.

Business Insight

These records represent discounted transactions with relatively high purchase amounts.

Recommendation

Compare these transactions with similar non-discounted records. Evaluate whether discounts are associated with additional purchases, higher conversion, or reduced margins before expanding promotional campaigns.

6.3 Top Five Products by Average Customer Rating

RANK	PRODUCT	AVERAGE RATING
1	Gloves	3.86/5
2	Sandals	3.84/5
3	Boots	3.82/5
4	Hat	3.80/5
5	Skirt	3.78/5

Business Insight

These products have the highest reported average ratings in the supplied SQL results. The differences between ratings are relatively small.

Recommendation

Review customer feedback for these products and investigate whether their positive ratings are associated with stronger sales performance. Ratings alone do not establish profitability or statistical significance.

6.4 Average Purchase Amount by Shipping Type

SHIPPING TYPE	AVERAGE PURCHASE AMOUNT
EXPRESS	$60.48
STANDARD	$58.46
DIFFERENCE	**$2.02**

The average purchase amount for Express-shipping transactions is approximately 3.46% higher than the Standard-shipping average.

Business Insight

Express-shipping transactions are associated with higher average purchase amounts in the reported results. This does not prove that Express shipping causes higher spending.

Recommendation

Test Express-shipping offers for selected customer groups and compare purchase amounts, shipping costs, conversion, and contribution margin.

6.5 Subscription Status Analysis

Reported SQL Results

SUBSCRIPTION STATUS	REPORTED RECORDS	AVERAGE PURCHASE AMOUNT	REPORTED TOTAL
NO	2,847	$59.87	$170,436
YES	1,053	$59.49	$62,645
TOTAL	**3,900**	—	**$233,081**

Business Insight

The reported SQL results indicate a slightly higher average purchase amount for the No group. However, these totals conflict with the earlier Power BI summary, which attributed the larger recorded purchase value to subscribers.

Recommendation

Reconcile the SQL and Power BI measures before drawing conclusions about subscription performance. Compare average purchase amount, repeat-purchase rate, and retention only after the underlying results have been verified.

6.6 Products with the Highest Discount Rates

RANK	PRODUCT	DISCOUNT RATE
1	Hat	50.00%
2	Sneakers	49.66%
3	Coat	49.07%
4	Sweater	48.17%
5	Pants	47.37%

Business Insight

These products have relatively high reported discount rates and are candidates for a closer review of promotional activity.

Recommendation

Compare discount rates with purchase volume, average purchase amount, inventory levels, and gross margin. Consider testing lower discounts to assess whether sales performance can be maintained while improving margins.

6.7 Customer Segmentation

Results

SEGMENT	NUMBER OF RECORDS	SHARE OF RECORDS
LOYAL	3,116	79.90%
RETURNING	701	17.97%
NEW	83	2.13%
TOTAL	**3,900**	**100.00%**

Business Insight

The supplied segmentation assigns most records to the Loyal group, followed by Returning and New.

These figures should be treated as record counts until the uniqueness of customer identifiers and the segmentation rules have been verified.

Recommendation

After validating the segments, develop retention campaigns for loyal customers, re-engagement campaigns for returning customers, and onboarding campaigns for new customers.

6.8 Top Products Within Each Category

Accessories

PRODUCT	PURCHASE RECORDS
JEWELRY	171
SUNGLASSES	161
BELT	161

Clothing

PRODUCT	PURCHASE RECORDS
BLOUSE	171
PANTS	171
SHIRT	169

Business Insight

These products have some of the highest purchase-record counts in their respective categories.

Recommendation

Monitor product availability and stock levels for high-volume items. Use margin, returns, and inventory data to decide which products should receive additional commercial attention.

The reported results for Footwear and Outerwear can be added if available.

6.9 Repeat Buyers by Subscription Status

Results

SUBSCRIPTION STATUS	REPEAT-BUYER RECORDS
NO	2,518
YES	958
TOTAL	**3,476**

These records meet the criterion of having more than five previous purchases.

Business Insight

The supplied results show more repeat-purchase records in the No subscription group than in the Yes group. Counts alone are insufficient to determine which group has the higher repeat-purchase rate because the sizes of the two subscription groups differ.

Recommendation

Compare repeat-purchase rates within each subscription group, validate customer-level uniqueness, and investigate whether subscription status is associated with retention or average purchase amount.

6.10 Purchase Value by Age Group

AGE GROUP	RECORDED PURCHASE VALUE	SHARE OF TOTAL
YOUNG ADULT	$62,143	26.66%
MIDDLE-AGED	$59,197	25.40%
ADULT	$55,978	24.02%
SENIOR	$55,763	23.93%
TOTAL	**$233,081**	**100.00%**

Business Insight

Young Adults account for the highest recorded purchase value, followed by Middle-aged customers.

Total purchase value may reflect differences in record counts as well as average spending behavior.

Recommendation

Compare average purchase amount, record count, purchase frequency, and product preferences by age group before allocating additional marketing budget.

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7. Power BI Dashboard Analysis

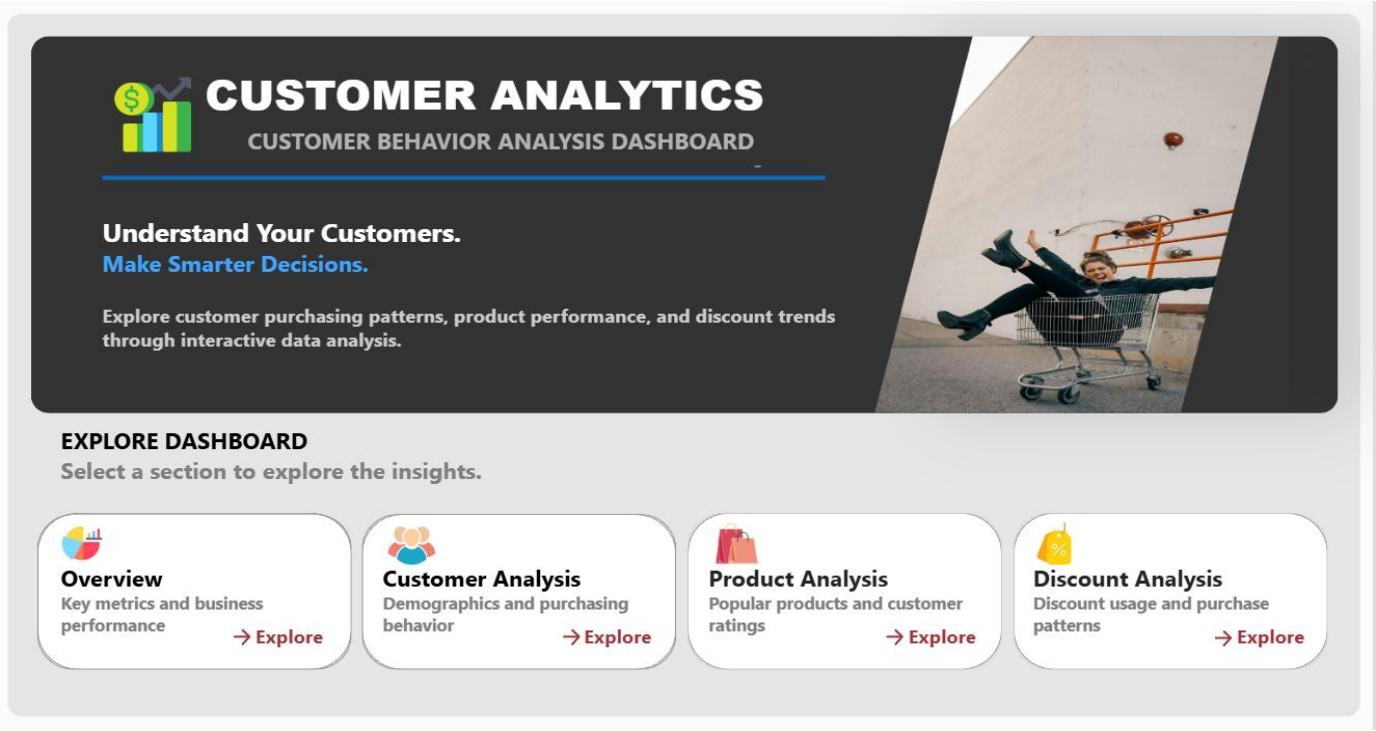
The Power BI report is organized into five pages: Landing Page, Overview, Customer Analysis, Product Analysis, and Discount Analysis.

7.1 Landing Page

The landing page provides navigation to the main analytical sections and serves as the entry point for stakeholders.

Business value: Clear navigation makes the report easier to use and allows decision-makers to access the relevant analysis quickly.

Figure 5 — Landing Page



Caption: Navigation interface for the customer behavior analytics report.

7.2 Overview Page

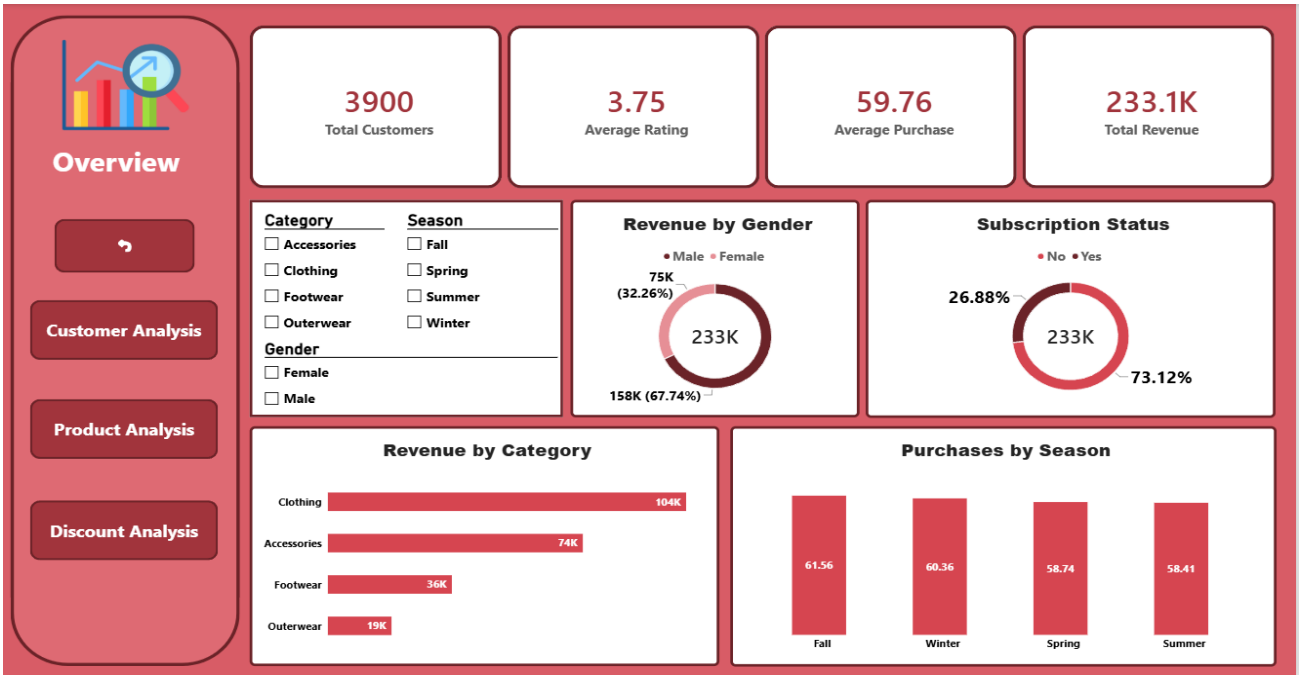
Key Performance Indicators

KPI	REPORTED VALUE
PURCHASE RECORDS	3,900
TOTAL RECORDED PURCHASE VALUE	$233,081
AVERAGE PURCHASE AMOUNT	$59.76
AVERAGE REVIEW RATING	3.75/5
DISCOUNT RATE	43.0%
DISCOUNTED RECORDS	1,677

Business Insight

The Overview page provides a high-level view of purchasing activity and discount usage. It establishes the baseline for more detailed analysis of customer behavior, product performance, and promotional patterns.

Figure 6 — Overview Dashboard



Caption: Executive overview of purchase value, average purchase amount, ratings, and discount activity.

7.3 Customer Analysis Page

Age Group Distribution

AGE GROUP	NUMBER OF RECORDS
YOUNG ADULT	1,028
MIDDLE-AGED	986
SENIOR	944
ADULT	942
TOTAL	**3,900**

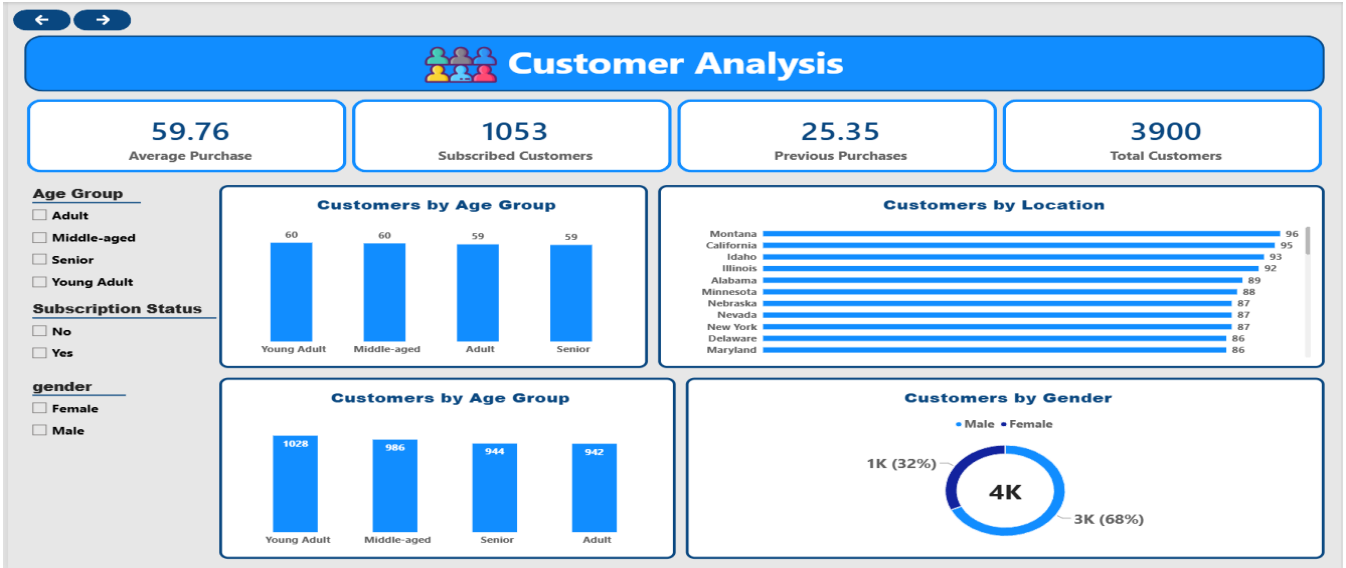
Additional reported metrics include an average of 25.35 previous purchases and recorded purchase value of $157,890 for Male records versus $75,191 for Female records.

Business Insight

The age groups have relatively similar record counts. Young Adults contribute the highest total recorded purchase value, but further comparisons are required to distinguish group size from spending behavior.

Subscription comparisons should not be used until the discrepancies between SQL and Power BI have been resolved.

Figure 7 — Customer Analysis Dashboard



Caption: Demographic distribution, purchasing history, and customer-related metrics.

7.4 Product Analysis Page

Purchase Records by Category

CATEGORY	PURCHASE RECORDS	SHARE
CLOTHING	1,737	44.54%
ACCESSORIES	1,240	31.79%
FOOTWEAR	599	15.36%
OUTERWEAR	324	8.31%
TOTAL	**3,900**	**100.00%**

Clothing and Accessories together represent approximately 76.33% of purchase records.

Leading Products by Purchase-Record Count

PRODUCT	PURCHASE RECORDS
BLOUSE	171
JEWELRY	171
PANTS	171
SHIRT	169
DRESS	166
SWEATER	164
JACKET	163
BELT	161
COAT	161
SUNGLASSES	161

Leading Products by Recorded Purchase Value

PRODUCT	APPROXIMATE VALUE
BLOUSE	$10.4K
SHIRT	$10.3K
DRESS	$10.3K
PANTS	$10.1K
JEWELRY	$10.0K
SUNGLASSES	$9.6K
BELT	$9.6K
SCARF	$9.6K
SWEATER	$9.5K
SHORTS	$9.4K

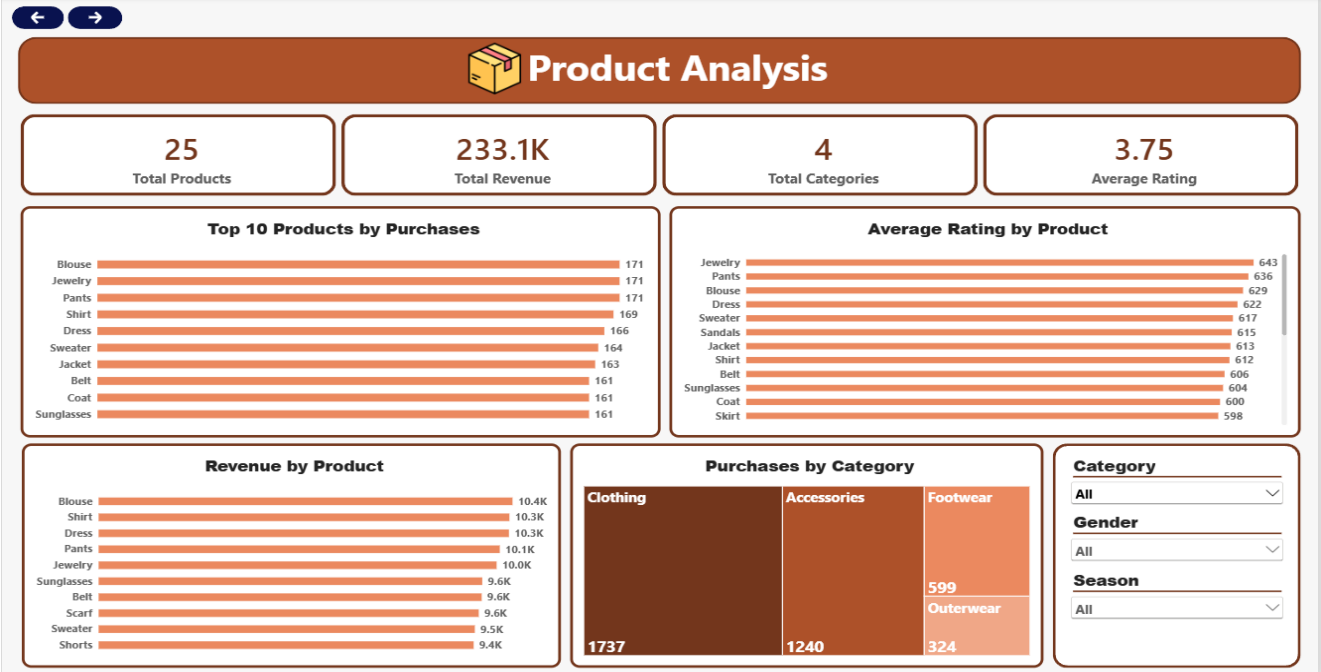
Business Insight

Clothing is the largest category by purchase-record volume. Several products appear among the leaders by both record count and recorded purchase value.

Recommendation

Use product-level demand and inventory data to prioritize availability. Incorporate costs and gross margins before treating high recorded value as evidence of high profitability.

Figure 8 — Product Analysis Dashboard



Caption: Category performance, popular products, and recorded purchase value by product.

7.5 Discount Analysis Page

Discount Summary

METRIC	REPORTED VALUE
DISCOUNTED RECORDS	1,677
DISCOUNT RATE	43.0%
AVERAGE PURCHASE AMOUNT WITH DISCOUNT	$59.28
AVERAGE PURCHASE AMOUNT WITHOUT DISCOUNT	$60.13
DIFFERENCE	$0.85

Discounted Records by Category

CATEGORY	DISCOUNTED RECORDS
CLOTHING	731
ACCESSORIES	543
FOOTWEAR	259
OUTERWEAR	144
TOTAL	**1,677**

Business Insight

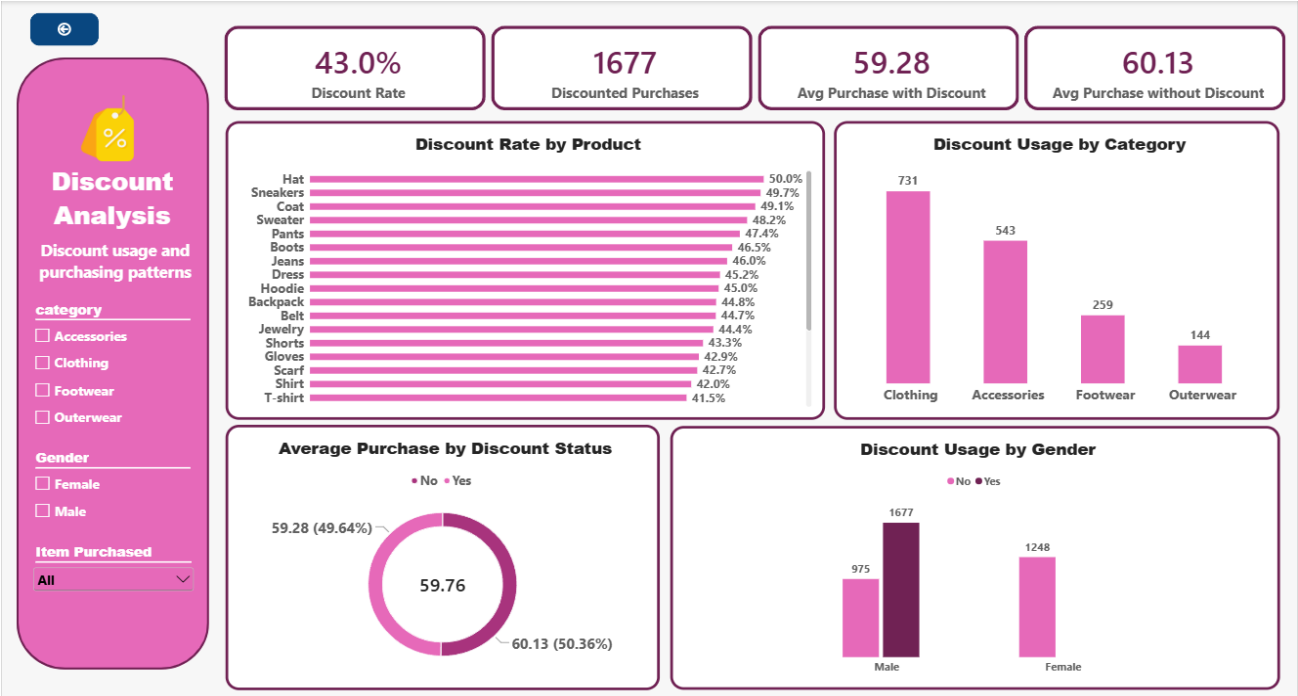
The average purchase amount without a discount is $0.85 higher than the average for discounted records. This is an observed difference, not proof that discounts reduce spending.

The reported gender breakdown of discounted records requires validation because all discounted records appear assigned to Male, with none assigned to Female.

Recommendation

Validate the gender breakdown and the dashboard calculations. Then evaluate promotional effectiveness using average purchase amount, conversion, and gross margin.

Figure 9 — Discount Analysis Dashboard



Caption: Discount utilization, average purchase comparisons, and product-level discount patterns.

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8. Key Business Findings

8.1 Category Concentration

Clothing and Accessories account for 76.33% of purchase records.

Opportunity: Improve availability monitoring and category-level performance reviews for these categories.

8.2 Demographic Differences

Male-associated records account for 67.74% of recorded purchase value, while Young Adults contribute the highest recorded value among age groups.

Opportunity: Investigate differences in average purchase amount, product preferences, and purchase frequency before designing targeted campaigns.

8.3 Discount Utilization

A discount was applied in 1,677 records, representing 43.0% of all records.

Opportunity: Assess whether discounts produce incremental purchases and whether the associated sales value justifies the discount cost.

8.4 Shipping Differences

Express-shipping transactions have a $60.48 average purchase amount, compared with $58.46 for Standard shipping.

Opportunity: Test selected Express-shipping offers while monitoring shipping costs and contribution margin.

8.5 Product Popularity

Several products have purchase-record counts between 161 and 171, and Clothing is the largest category.

Opportunity: Use demand and stock availability data to improve inventory planning.

8.6 Customer Segmentation

The reported segmentation identifies 3,116 Loyal, 701 Returning, and 83 New records.

Opportunity: Use validated segments to develop retention, re-engagement, and onboarding campaigns.

8.7 Reporting Reliability

The subscription results conflict between SQL and Power BI, and the discount-by-gender visualization needs further investigation.

Opportunity: Resolve these discrepancies before using the affected metrics in management decisions.

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9. Recommendations and Action Plan

Priority 1: Improve Data Reliability

Actions

- Reconcile subscription figures between SQL and Power BI.
- Validate the discount-by-gender visualization.
- Confirm whether customer identifiers are unique.
- Verify the average-rating measure and scale.
- Document KPI definitions.

Success measures

- SQL and Power BI results agree under the same filters.
- Discount classifications are verified.
- Customer-level measures use a validated unit of analysis.

Priority 2: Improve Product and Inventory Decisions

Actions

- Monitor frequently recorded products.
- Compare product volume and recorded purchase value.
- Incorporate inventory, costs, returns, and gross margins when available.

Success measures

- Product availability.
- Stockout rate.
- Sell-through rate.
- Gross margin by product and category.

Priority 3: Evaluate Discounts More Effectively

Actions

- Compare discounted and non-discounted purchases by product.
- Review high-discount products.
- Test selected discount levels.
- Measure margin and conversion alongside purchase amount.

Success measures

- Conversion rate.
- Average purchase amount.
- Gross margin per transaction.
- Incremental sales against a comparison group.

Priority 4: Strengthen Customer Retention

Actions

- Validate customer segment definitions.
- Design retention campaigns for established customers.
- Test onboarding campaigns for new customers.
- Compare subscription groups after reconciling the underlying data.

Success measures

- Repeat-purchase rate.
- Customer retention rate.
- Time between purchases.
- Subscription conversion rate.

Priority 5: Evaluate Shipping Strategy

Actions

- Compare purchase amounts by shipping type.
- Include shipping costs and delivery performance.
- Test Express shipping offers for selected customer groups.

Success measures

- Express-shipping adoption.
- Delivery cost per order.
- On-time delivery.
- Customer satisfaction.
- Contribution margin.

10. Management KPI Scorecard

The following scorecard is recommended for ongoing business monitoring.

KPI	BUSINESS PURPOSE
TOTAL RECORDED PURCHASE VALUE	Monitor the total value of recorded purchases
AVERAGE PURCHASE AMOUNT	Track average transaction value
DISCOUNT RATE	Monitor the frequency of discount use
AVERAGE PURCHASE AMOUNT BY DISCOUNT STATUS	Compare discounted and non-discounted transactions
AVERAGE REVIEW RATING	Monitor customer feedback
CATEGORY PURCHASE SHARE	Understand the distribution of purchase records
PRODUCT PURCHASE COUNT	Support demand and stock reviews
REPEAT-PURCHASE RATE	Monitor repeat behavior
CUSTOMER RETENTION RATE	Track customer retention over time
GROSS MARGIN	Evaluate profitability
STOCKOUT RATE	Monitor inventory availability
EXPRESS SHIPPING ADOPTION	Understand shipping preferences

Gross margin, retention, and stockout measures require additional data or validated customer-level and time-based records.

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11. Limitations and Data Validation

11.1 Purchase Records Are Not Necessarily Unique Customers

The dataset contains 3,900 records, but the number of distinct customers has not been independently established. Customer-level claims require validation of the customer identifier and data structure.

11.2 Recorded Purchase Value Is Not Necessarily Net Revenue

The sum of purchase amounts is reported as recorded purchase value. Without verified accounting definitions, costs, returns, taxes, and other adjustments, it should not automatically be described as net revenue or profit.

11.3 Observed Differences Do Not Prove Causation

Differences across discount and shipping groups do not establish that discounts or shipping options caused changes in purchase amounts.

11.4 Subscription Results Require Reconciliation

The SQL output reports $170,436 for the No group and $62,645 for the Yes group. The earlier Power BI summary attributed the larger total to subscribers. This discrepancy must be resolved before the results are used for business decisions.

11.5 Discount-by-Gender Results Require Validation

The reported visualization shows all discounted records assigned to Male and none to Female. This should be treated as an unresolved reporting or data-quality issue until verified.

11.6 Product Rating Measure Requires Validation

The reported SQL top-five ratings are between 3.78 and 3.86 out of 5. Any separate visual displaying much larger values under an average-rating label should be checked to confirm that it is calculating average ratings rather than record counts.

11.7 Missing Sales-Channel Information

The available fields do not clearly identify online and offline transactions. A channel-level comparison requires additional data.

11.8 Additional Data That Would Improve the Analysis

- Product costs and gross margins.
- Returns and cancellations.
- Transaction dates and customer-level transaction history.
- Inventory and stock availability.
- Sales channel.
- Shipping costs and delivery performance.
- Validated customer identifiers and subscription history.

These fields would support stronger profitability, retention, and promotional-effectiveness analysis.

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12. Conclusion

This project demonstrates an end-to-end approach to retail data analytics, combining data preparation, SQL analysis, database integration, and interactive Power BI reporting.

The analysis identifies patterns in product-category volume, recorded purchase value, demographic groups, discount usage, shipping options, customer ratings, and purchasing history.

The findings highlight opportunities to improve inventory planning, evaluate promotional strategies, test shipping offers, and develop more targeted customer engagement initiatives.

The project also identifies discrepancies requiring further validation. Reconciling subscription measures, checking the discount-by-gender visualization, and confirming the customer-level unit of analysis are important steps toward reliable management reporting.

The main outcome is a structured analytical workflow that transforms raw retail data into business insights, identifies reporting risks, and proposes measurable actions for further evaluation.

13. Screenshot Placement Guide

Use screenshots from the actual project. Keep dashboard screenshots large enough to read and avoid including passwords or sensitive information.

FIGURE	SCREENSHOT	LOCATION
1	Dataset structure and columns	Section 3
2	Missing-value assessment and preparation results	Section 4
3	End-to-end analytics workflow	Section 5
4	Prepared data in PostgreSQL	Section 5
5	Power BI Landing Page	Section 7.1
6	Power BI Overview	Section 7.2
7	Power BI Customer Analysis	Section 7.3
8	Power BI Product Analysis	Section 7.4
9	Power BI Discount Analysis	Section 7.5

Final Quality Checklist

- Ensure monetary values are labelled in USD.
- Use actual screenshots from the project.
- Reconcile subscription figures before final submission.
- Validate the discount-by-gender visualization.
- Confirm that rating charts display average ratings correctly.
- Distinguish purchase records from unique customers.
- Avoid claiming increased revenue or profit without measured evidence.
- Keep the final report focused on business questions, numerical findings, recommendations, and measurable KPIs.